

Task on EC2

- 1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.**

1. Launch EC2 instance.
2. Select Amazon Linux 2 AMI.
3. Select instance type, key pair, subnet and security group.
4. In Advanced details → User data, paste the above script.
5. Launch the instance.
6. Connect using SSH.
7. Check Apache

```
USER@DESKTOP-NTBFNMK MINGW64 ~/Downloads
$ ssh -i "pop.pem" ec2-user@ec2-100-58-127-29.compute-1.amazonaws.com
The authenticity of host 'ec2-100-58-127-29.compute-1.amazonaws.com (100.58.127.29)' can't be established.
ED25519 key fingerprint is SHA256:/VTOwH04Dz6d8GwJw/CGDYvWGe4V6b1Fnjjhdw/kpqQ.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-100-58-127-29.compute-1.amazonaws.com' (ED25519) to the list of known hosts

#--
#####
#####
###|
#|
V-+> https://aws.amazon.com/linux/amazon-linux-2023
m/_/
last login: Wed Sep 30 10:28:52 2026 from 18.206.107.28
[ec2-user@ip-172-31-27-216 ~]$ sudo su -
[root@ip-172-31-27-216 ~]# systemctl status httpd
● httpd.service The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-30 10:27:07 UTC; 23min ago
     Docs: man:httpd.service(8)
    Main PID: 4391 (httpd)
      Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec 0; Bytes served/sec: 0 B/sec"
        Tasks: 177 (limit: 1113)
       Memory: 13.5M
         CPU: 971ms
    CGroup: /system.slice/httpd.service
            └─4391 /usr/sbin/httpd -DFOREGROUND
              4441 /usr/sbin/httpd -DFOREGROUND
              4447 /usr/sbin/httpd -DFOREGROUND
              4448 /usr/sbin/httpd -DFOREGROUND
              4459 /usr/sbin/httpd -DFOREGROUND

Sep 30 10:27:07 ip-172-31-27-216.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server..
Sep 30 10:27:07 ip-172-31-27-216.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 30 10:27:07 ip-172-31-27-216.ec2.internal httpd[4391]: Server configured, listening on: port 80
[root@ip-172-31-27-216 ~]# ps -ef | grep httpd
root      4391          1  0 10:27 ?           00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    4441          1  0 10:27 ?           00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    4447          1  0 10:27 ?           00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    4448          1  0 10:27 ?           00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    4459          1  0 10:27 ?           00:00:00 /usr/sbin/httpd -DFOREGROUND
root      28286        2738    0 10:52 pts/1      00:00:00 grep --color=auto httpd
```

2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

User data - *optional* | [Info](#)

Upload a file with your user data or enter it in the field.

 Choose file

```
#!/bin/bash
```

```
apt update -y
apt install nginx -y
systemctl enable nginx
systemctl start nginx
```

```
USER@DESKTOP-NTBFNKM MINGW64 ~/Downloads
$ ssh -i test.pem ubuntu@54.159.131.166
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 30 11:36:38 UTC 2026

System load:  0.1          Processes:    116
Usage of /:   34.1% of 6.61GB   Users logged in:  0
Memory usage: 25%          IPv4 address for enX0: 172.31.31.147
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

191 updates can be applied immediately.
147 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-31-147:~$ nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
ubuntu@ip-172-31-31-147:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded /usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 11:31:38 UTC; 6min ago
     Invocation: 414e568e129e48bdaa96f1147335624c
       Docs: man:nginx(8)
    Main PID: 1484 (nginx)
      Tasks: 2 (limit: 657)
     Memory: 2.3M (peak: 5.1M)
        CPU: 27ms
     CGroup: /system.slice/nginx.service
             └─1484 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─1487 "nginx: worker process"
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

3. Launch one Windows server and install Tomcat on Windows.

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Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base

ami-00d8aa800578d8b12 (64-bit (x86))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

```

Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> java --version
openjdk 21.0.12.1 2026-08-18 LTS
OpenJDK Runtime Environment Microsoft-14941484 (build 21.0.12.1+1-LTS)
OpenJDK 64-Bit Server VM Microsoft-14941484 (build 21.0.12.1+1-LTS, mixed mode, sharing)
PS C:\Windows\system32> Expand-Archive -Path "C:\Users\Administrator\Downloads\apache-tomcat-10.1.60.zip" -DestinationPath "C:\Tomcat"
PS C:\Windows\system32> dir C:\Tomcat

Directory: C:\Tomcat

Mode                LastWriteTime         Length Name
----                -
d-----          10/2/2026  11:32 AM                apache-tomcat-10.1.60

PS C:\Windows\system32> cd C:\Tomcat\apache-tomcat-10.1.60\bin
PS C:\Tomcat\apache-tomcat-10.1.60\bin> java --version
openjdk 21.0.12.1 2026-08-18 LTS
OpenJDK Runtime Environment Microsoft-14941484 (build 21.0.12.1+1-LTS)
OpenJDK 64-Bit Server VM Microsoft-14941484 (build 21.0.12.1+1-LTS, mixed mode, sharing)
PS C:\Tomcat\apache-tomcat-10.1.60\bin> .\startup.bat
Using CATALINA_BASE:   "C:\Tomcat\apache-tomcat-10.1.60"
Using CATALINA_HOME:   "C:\Tomcat\apache-tomcat-10.1.60"
Using CATALINA_TMPDIR: "C:\Tomcat\apache-tomcat-10.1.60\temp"
Using JRE_HOME:        "C:\Program Files\Microsoft\jdk-21.0.12.101-hotspot\"
Using CLASSPATH:       "C:\Tomcat\apache-tomcat-10.1.60\bin\bootstrap.jar;C:\Tomcat\apache-tomcat-10.1.60\bin\tomcat-juli.jar"
Using CATALINA_OPTS:   ""
PS C:\Tomcat\apache-tomcat-10.1.60\bin>

```

```

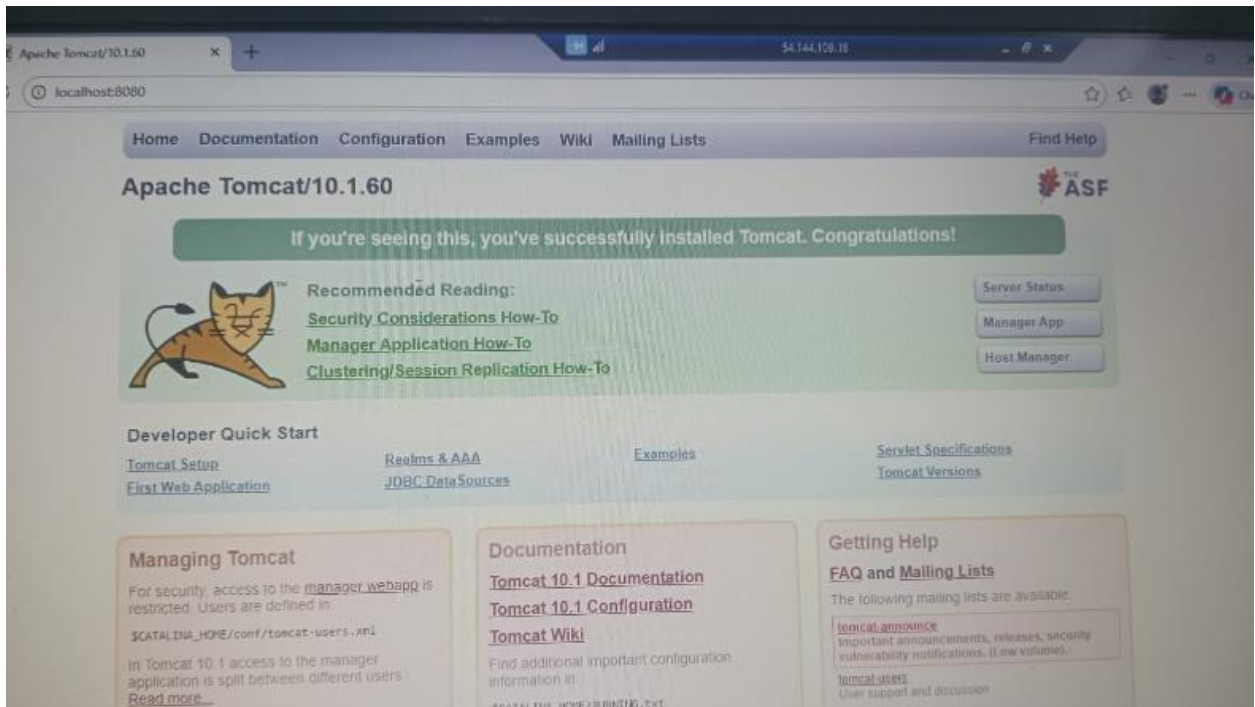
OpenJDK Runtime Environment Microsoft-14941484 (build 21.0.12.1+1-LTS)
OpenJDK 64-Bit Server VM Microsoft-14941484 (build 21.0.12.1+1-LTS, mixed
PS C:\Windows\system32> Expand-Archive -Path "C:\Users\Administrator\Down
PS C:\Windows\system32> dir C:\Tomcat

Directory: C:\Tomcat

Mode                LastWriteTime         Length Name
----                -
d-----          10/2/2026  11:32 AM                apache-tomcat-10.1.60

PS C:\Windows\system32> cd C:\Tomcat\apache-tomcat-10.1.60\bin
PS C:\Tomcat\apache-tomcat-10.1.60\bin> java --version
openjdk 21.0.12.1 2026-08-18 LTS
OpenJDK Runtime Environment Microsoft-14941484 (build 21.0.12.1+1-LTS)
OpenJDK 64-Bit Server VM Microsoft-14941484 (build 21.0.12.1+1-LTS, mixed
PS C:\Tomcat\apache-tomcat-10.1.60\bin> .\startup.bat

```



4. Take a snapshot of the instance created in Task 1.

Purpose: To create a backup of the EC2 instance's EBS volume.

- 1. Go to AWS Console → EC2 → Instances.**
- 2. Select the Task 1 Ubuntu instance.**
- 3. Open Storage → Root EBS Volume.**
- 4. Click Actions → Create snapshot.**
- 5. Enter description: Task-1-Ubuntu-Nginx-Snapshot**
- 6. Click Create snapshot.**
- 7. Go to EC2 → Snapshots and verify Status: Completed.**

rch [Alt+S] United States (N. Virginia) Parv

Successfully created snapshot snap-0b602ee785892b5ad from volume vol-09152f2b1ebaa4bc9. If you need your snapshot to be immediately available consider using Fast Snapshot Restore. Manage fast snapshot restore

Volumes (1) Info Last updated less than a minute ago Recycle Bin Actions Create volume

Saved filter sets Choose filter set Search

Volume ID = vol-09152f2b1ebaa4bc9 Clear filters

Name Volume ID Type Size IOPS Throughput Owner ID Snaos

Fault tolerance for all volumes in this Region

Snapshot summary Last updated on Thu, Oct 01, 2026, 03:21:14 PM (GMT+05:30)

Snapshots to back up volumes can be manually created, or created through automation such as Amazon Data Lifecycle Manager, AWS Backup, custom

UltraServers New

Capacity Manager New

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Volumes

Snapshots

Snapshots (1) Info Last updated 1 minute ago Recycle Bin Actions Create snapshot

Snapshot scope Owned by me Search

	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier
☐		snap-0b602ee785892b5ad	8 GiB	8 GiB	Task-1-Ubuntu-Nginx-Sna...	Standard

5. Assign passwordless authentication for the EC2 created in Task 2.

```
authorized_keys 1d_ed25519 1d_ed25519.pub
ubuntu@ip-172-31-44-65:~/.ssh$ cd ..
ubuntu@ip-172-31-44-65:~$ sudo find / -name authorized_keys
ubuntu@ip-172-31-44-65:~$ sudo find / -name authorized_keys
/home/ubuntu/.ssh/authorized_keys
/root/.ssh/authorized_keys
ubuntu@ip-172-31-44-65:~$ nano /home/ubuntu/.ssh/authorized_keys
ubuntu@ip-172-31-44-65:~$ cat /home/ubuntu/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAxQkvWgyhPmEH62jy+S8YF9PsZ1QdkXwt3cZULceECts0hQGgv1lqC
Jvu9HyiAP0KQry/RhqlbdUIamStv3ySdHmcirVYC3YYNp74Bs4qnx+dazlGipCC4M1/wRey3JV60eXtUZa3wizINL7b
t
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDmunBswg2QG/iWLMOA+tXQJRxRq53Rf3tNkFM0dowvicynKcT9gFH1
GVpgWyiEInw03GECNVEz70U87CzPeMopd0e1AxA9ibqnJMpuqvwKdVSGfpPMiLZAHKsWl8ZUVKP1b1cE7uh0j0etJdBoq
17aFLqsaSLB1wf0JRfbr0Vh7TEiGiVcIaVwLmLI03shHfncbH+2A1Z6Wpp1a05X+fnONOC+YMNUN6ovDNM5NcVh1fD6zE
ubuntu@ip-172-31-44-65:~$ exit
logout
Connection to 100.30.183.38 closed.

USER@DESKTOP-NTBFNKM MINGW64 ~/Downloads
$ ssh ubuntu@100.30.183.38
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Oct  1 11:00:37 UTC 2026

System load:  0.29                Processes:           115
Usage of /:   34.5% of 6.61GB     Users logged in:    0
Memory usage: 30%                IPv4 address for enX0: 172.31.44.65
```

6. Launch any EC2 using the spot purchasing option.

- 1) Go to EC2 → Instances.
- 2) Click Launch instance.
- 3) Scroll to Advanced details.
- 4) Find Purchasing option.
- 5) Select Spot Instances .
- 6) Configure the instance and click Launch instance.

EBS-optimized instance | [Info](#)

Disable

Instance bandwidth configuration | [Info](#)

Select

Purchasing option | [Info](#)

- ☐ None
If you don't select a purchasing option, your instance launches as an On-Demand Instance by default
- ☐ Capacity Blocks
Launch instances for your active capacity blocks
- ☐ Interruptible Capacity Reservations
Launch instances for your interruptible Capacity Reservations
- ☒ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

.

Successfully initiated stopping of i-068adb6c77542fd6

Instances (1/2) [Info](#)

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

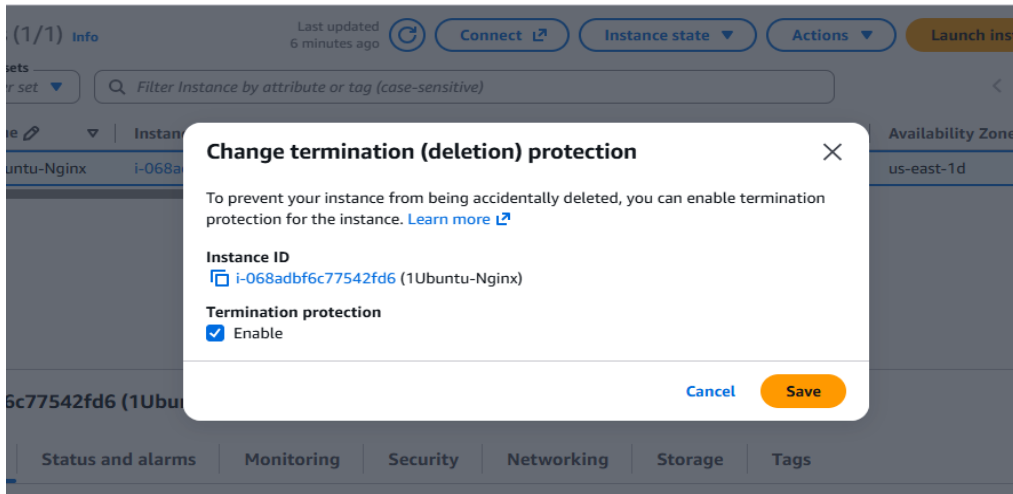
Choose filter set

Filter Instance by attribute or tag (case-sensitive)

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IP
☒	1Ubuntu-Nginx	i-068adb6c77542fd6	Stopping	t2.micro	–	us-east-1d	ec2-1
☐	Spot-EC2	i-0dcb4bd22ec37994	Running	t2.micro	Initializing	us-east-1d	ec2-3

7. Enable termination policy on the EC2 created in Task 2.



- ✓ EC2 instance selected
- ✓ Instance settings opened
- ✓ Termination protection enabled
- ✓ Saved successfully.

8. Launch one EC2 using AWS CLI.

```
PS C:\Users\USER> Test-Path "C:\Program Files\Amazon\AWSCLIV2\aws.exe"
True
PS C:\Users\USER> Test-Path "C:\Program Files\Amazon\AWSCLIV2\aws.exe"
True
PS C:\Users\USER> $env:Path += ";C:\Program Files\Amazon\AWSCLIV2"
PS C:\Users\USER> aws --version
aws-cli/2.37.8 Python/3.14.6 Windows/11 exe/AMD64
PS C:\Users\USER> |
```